# Akash Saini

+91 9636100337 aksainibits@gmail.com LinkedIn Delhi

## EDUCATION

**Birla Institute of Technology and Science, Pilani (Pilani Campus)** *Aug 2019 – May 2023*
Integrated Master of Science in Mathematics **CGPA: 6.49**

**Relevant Coursework:** Machine Learning, Artificial Intelligence, Foundations of Data Science, Optimization, Probability and Statistics, Information Retrieval, Linear Algebra, Applied Statistical Methods

## CORE COMPETENCIES

**AI & Frameworks:** Python, PyTorch, TensorFlow, Transformers (Hugging Face), scikit-learn
**LLM & Agentic AI:** RAG Pipelines, Vector DBs (Chroma, FAISS), Prompt Engineering
**MLOps & Infrastructure:** Docker, Kubernetes, GitHub Actions.
**Cloud & Data:** PostgreSQL, MySQL, Linux, SQL, Pandas, NumPy

## EXPERIENCE

**Learning Facilitator — WILP, BITS Pilani** *Aug 2025 – Present*
- **Collaborate with faculty and technical teams** to manage digital platforms and facilitate core AI-foundation courses, including **Computer Programming, Linear Algebra, and Optimization**.

**AI/LLM Training Workflows - Senior Associate — Innodata** *Jan 2026 – June 2026*
- Curated and audited large-scale textual datasets for the **instruction fine-tuning of Gemini models**, ensuring high response accuracy, context relevance, and strict compliance with AI safety guidelines.
- Applied **prompt engineering** techniques to rigorously evaluate LLM outputs, identifying edge cases and mitigating model hallucinations to improve foundational NLP capabilities.

**Software Development Intern — PCHF** *Jul 2022 – Dec 2022*
- **Resolved 40+ bugs** to enhance reliability, wrote Java unit tests for increased coverage.
- Developed and implemented a validation check to prevent malicious file uploads, ensuring data security.
- Designed and deployed APIs to update loan status and created Python/MongoDB scripts for database.
- Creating custom code aligned with business requirements, ensuring compliance and operational efficiency.
- **Investigated root causes** of token expiry, implemented fixes and alerts for service downtime detection.
- Performed root cause analysis for duplicates, developed checks ensuring unique customer record creation.
- Collaborated in an Agile environment, following SCRUM practices with end-to-end ownership of features.

**Machine Learning Engineer Intern — Swecha** *May 2021 – Jul 2021*
- Engineered automated **data ingestion and pre-processing pipelines** in Python, transforming unstructured text to scale the training database by over 6000x.
- **Led a 10-member team** to develop "Swecha Voice," an open-source, NLP-driven speech recognition AI system leveraging Python frameworks.

## AI & ML PROJECTS

**Fake News Detection Pipeline — NLP & Text Classification** *Jan 2023 – May 2023*
- Engineered an **NLP data pipeline** using NLTK for pre-processing, tokenization, and feature engineering.
- Optimized classification models via hyperparameter tuning to achieve **95% accuracy** in text evaluation.

**Statistical Forecasting of Energy Data — Time Series Modeling** *Feb 2022 – June 2022*
- Engineered a comparative **time-series forecasting** pipeline to analyze 15 years of state energy data, executing extensive data cleaning and exploratory data analysis (EDA).
- Evaluated predictive models (AR, MA, ARIMA, SARIMA) using RMSE and MAPE evaluation metrics, successfully identifying the optimal architecture for high-accuracy forecasting.

## LEADERSHIP & CROSS-FUNCTIONAL EXPERIENCE

**Head of Data Science Team — Mathematics Association** *Aug 2021 – Jun 2022*
- **Founded and led a 10-member Data Science team**, directing technical strategy and driving analytics-focused initiatives to foster a data-driven environment.
- Mentored members in building end-to-end data analysis pipelines and authored technical documentation to promote internal knowledge sharing on complex datasets.

**Head of Operations — Nirmaan Organization** *Aug 2019 – Jun 2023*
- Directed operations and cross-functional collaboration for a non-profit, **leading 100+ volunteers** to successfully execute large-scale projects in dynamic environments.